Southern Africa
Small Business Markets

Local
Shops

Informal
Markets

Financial Health Index for Small Businesses

Predicting Resilience Beyond Profit

Eswatini

Lesotho

Malawi

Zimbabwe

Problem Statement

Many small businesses across Southern Africa are financially fragile — yet invisible to banks, governments, and support systems.

43M+

SMEs excluded from formal credit across Southern Africa

60%

of small businesses fail within 5 years due to financial shocks

0

data-driven holistic tools exist to measure true SME health

Why Profit Isn't Enough

- Banks and governments only focus on profit — a snapshot of today
- Profit doesn't show if businesses can survive crises like illness or drought
- Small businesses are the backbone of the economy in all 4 countries

Profit vs Resilience

Profit (Today)	Resilience (Long-Term)
✗ Shows if you made money now	✓ Shows if you can survive crises
✗ Snapshot only — one moment	✓ Holistic: Savings, Debt, Recovery
✗ Ignores emergencies	✓ Helps plan for Rainy Days

We need BOTH — and machine learning delivers it.

The Solution: Financial Health Index (FHI)

- **Holistic health check** — Goes beyond income to measure real stability
- **Measures Resilience** — Not just profit — actual survival ability
- **Savings** — Safety net available when crises hit?
- **Debt** — Can they manage what they owe?
- **Recovery** — Can they bounce back from a crisis?

Financial Health Index (FHI)

01
Savings

Do they have a safety net?

02
Debt

Can they manage what they owe?

03
Recovery

Can they bounce back from a crisis?

What We Want to Achieve

USE MACHINE LEARNING TO PREDICT BUSINESS FINANCIAL HEALTH

Using survey data from 4 nations

CATEGORIZE BUSINESSES INTO: LOW, MEDIUM, OR HIGH HEALTH

3-class classification output

Output Classifications:

LOW HEALTH

Struggling, No Savings, High Debt

MEDIUM HEALTH

Doing Okay, but Vulnerable

HIGH HEALTH

Strong, Safe, Able to Grow

Community Business Owners

Informal Market Traders

How We Do It

Using Data Clues

Who They Are

Age, gender, education, location, household size

What They Do

Local sales, import/export activity, traded goods

Financial Habits

Banking, mobile money, loans, insurance held

Business Size

Employees vs solo, turnover, expenses, age of firm

Governments Target Aid

Banks Lend Responsibly

Businesses Survive & Grow

Why This Matters

Governments have limited money

They can't help everyone — this tool pinpoints who needs support most.

Banks are afraid to lend

Small shops don't have fancy paperwork — FHI gives them a credible risk score.

It's about the "Rainy Day"

Businesses that survive shocks grow — this model identifies who will make it.

The Machine Learning Approach

01

Collect Data

Survey data from 4 Southern African countries — demographics, finances, attitudes

02

Engineer Features

Create smart indicators like Financial Access Score, Profit Proxy, Insurance Score

03

Train Models

3 algorithms (Random Forest, Gradient Boosting, Extra Trees) vote together as an ensemble

04

Predict Health

Output: LOW, MEDIUM, or HIGH health for every business — with 89% accuracy

Our Results

89%

Accuracy
(Macro F1 Score)

Predicting Low / Medium / High
financial health correctly

91% **LOW Health**

Struggling businesses correctly flagged for urgent intervention

87% **MEDIUM Health**

Vulnerable businesses identified before they fall into crisis

88% **HIGH Health**

Resilient businesses recognised for credit and growth support

Tested on real data from Eswatini · Lesotho · Malawi · Zimbabwe

Call to Action / Next Steps

Moving Forward

- Deploy the FHI model in pilot regions across all 4 countries
- Collaborate with banks and government agencies on real-world rollout
- Track improvements in business resilience and financial inclusion over time

Our Vision for Southern Africa

1 Million SMEs

reach financial services they were excluded from

4 Countries

with tools to direct aid exactly where needed

Resilient Economies

built from the ground up, business by business

Global Template

replicable in any developing economy worldwide